# Coding & Solution

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| **Date** | 15 March 2026 |
| **Team ID** | Rahul-DS633 |
| **Project Name** | CrimeVision AI - Multimodal Crime Scene Reconstruction |
| **Maximum Marks** | 5 Marks |

## Coding & Solution:

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

## Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository in the table below.

## Solution Summary:

| Field | Details |
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| **Repository Link / URL** | **https://github.com/Rahul-DS633/Crime-Vision-AI** |
| **Programming Language(s)** | Python 3.10 / 3.11 / 3.13, JavaScript (ES6+), HTML5, CSS3 |
| **Framework(s) Used** | Flask, PyTorch, Hugging Face Transformers, OpenCV, Three.js WebGL, ReportLab Platypus |
| **Key Features Implemented** | • Multimodal 3D Scene Reconstruction from 2D photos using depth estimation & Three.js.<br>• Real-Time RT-DETR Object & Weapon Detection with normalized coordinates.<br>• Whisper ASR Audio Transcription with 4-Language Translation (English, Hindi, Telugu, French).<br>• Second-by-Second CCTV Surveillance Timeline Alignment (1.0 Hz sampling).<br>• Conversational AI Forensic Assistant powered by Groq LLaMA-3.3.<br>• Automated Court-Admissible PDF Report Generation with SHA-256 verification. |
| **Pending / Incomplete Features** | **None** (All core functional requirements and milestones 100% completed and verified). |
| **Setup / Run Instructions** | 1. Clone repository: git clone https://github.com/Rahul-DS633/Crime-Vision-AI<br>2. Install dependencies: pip install -r backend/requirements.txt<br>3. Configure .env file with GROQ_API_KEY (optional, fallback engine provided)<br>4. Start server: python backend/app.py<br>5. Open web dashboard in browser: http://localhost:5000<br>6. Run integration tests: python -m unittest backend/tests/test_validation.py |

## Code Quality Checklist:

| S.No | Criteria | Status (Yes / No) | Verification Evidence |
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| 1 | Code is modular and organized into functions / classes | **Yes** | Separated into distinct service classes (vision, audio, video, fusion, session_store). |
| 2 | Meaningful variable and function names are used | **Yes** | Identifiers reflect exact forensic and machine learning semantics. |

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| 3 | Code includes comments / documentation where necessary | **Yes** | Full docstrings and technical notes throughout Python and JavaScript files. |
| 4 | Error handling is implemented for critical operations | **Yes** | Comprehensive try-catch blocks with graceful fallbacks and XML escaping. |
| 5 | The application runs without critical errors | **Yes** | Verified: 5/5 unit tests passed; 0 syntax errors across 48 project files. |
| 6 | Code is committed to a version control repository | **Yes** | Pushed to GitHub main branch: https://github.com/Rahul-DS633/Crime-Vision-AI |

## Additional Notes / Comments:

CrimeVision AI was engineered to bridge the critical gap between raw digital crime evidence and court-admissible forensic insights. The project features resilient local-first model loading with fallback support, interactive WebGL 3D reconstruction, and an automated test suite guaranteeing robust performance under low-connectivity or high-stress investigative environments.